

D905

Statistics for Secondary Mathematics Teaching

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

At a planning session held in the school library, a teacher drafts questions for a data unit that starts after winter break. Which is a statistical question?

- A. How many hours of sleep do students at this school typically get on school nights?
- B. Any question whose answer is a number.
- C. How tall is the classroom door?
- D. How many students are enrolled in this class today?

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A class of 26 wonders whether students who play an instrument spend more time on homework. The school has a new music wing. What should the class do first?

- A. Make a histogram of homework hours.
- B. Collect survey data from students right away.
- C. Write a clear statistical question that defines the groups and how homework time is measured.
- D. Interpret the results and decide whether music helps.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

Heights of students measured with a new stadiometer in the nurse's office have a mean of 160 cm and a standard deviation of 8 cm. The measurements took three days. What is the z-score for a height of 176 cm?

- A. 22
- B. 16
- C. -2
- D. 2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Scores on a test are normal with mean 500 and standard deviation 100. What score has $z = -1.5$?

- A.** 498.5
- B.** 650
- C.** 150
- D.** 350

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

Two hospitals, one of which has a new parking garage, record births: the larger has about 45 births a day and the smaller about 15. Over a year, which hospital has more days on which over 60% of babies born are boys?

- A.** Neither; each day is always exactly 50% boys.
- B.** About the same, since the chance of a boy is the same at both.
- C.** The smaller hospital.
- D.** The larger hospital.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A fair coin lands heads 7 times in its first 10 flips. As the total number of flips grows to 10,000, what is expected to happen?

- A.** The proportion of heads moves toward 0.5.
- B.** The proportion stays near 0.7.
- C.** The count of heads minus tails must shrink toward 0.
- D.** Tails must come up more often to balance the early heads.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A principal wants a survey that represents grades 9 through 12, so she randomly selects 25 students from each grade. The survey was emailed on a Monday. What sampling method is this?

- A. Convenience sampling, because the survey was sent by email.
- B. Cluster sampling.
- C. Simple random sampling.
- D. Stratified random sampling.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A student newspaper staff of 12 polls 400 randomly chosen students about a later start time. The poll ran the week of the spring dance. What is the approximate 95% margin of error?

- A. About $\pm 0.25\%$.
- B. About $\pm 5\%$.
- C. About $\pm 2.5\%$.
- D. About $\pm 20\%$.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A pollster wants a 95% margin of error of about $\pm 2\%$ (an interval 4 points wide). About how many people should be sampled?

- A. 625
- B. 2,500
- C. 50
- D. 5,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

At a school carnival that raised \$900, a game pays out if a player rolls at least one 6 in four rolls of a fair die. The booth was run by parent volunteers. What is the probability of winning?

- A. About 0.482
- B. About 0.518
- C. About 0.667
- D. About 0.0008

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Three independent parts each have the same defect probability p . The chance that at least one is defective is 0.271. What is p ?

- A. 0.1
- B. 0.09
- C. 0.729
- D. 0.9

Target word _____	The one fact that decides _____	Answer __	☐ sure ☐ unsure ☐ guess
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12.

During an after-lunch lesson, a student says HTHTTH is more likely than HHHHHH in six flips of a fair coin. The coins are plastic classroom counters. What is the correct response?

- A. HHHHHH is impossible with a fair coin.
- B. HHHHHH is more likely because streaks tend to continue.
- C. Both sequences have probability $1/64$; the mixed one only looks more random.
- D. HTHTTH is more likely because it has three heads.

Target word _____	The one fact that decides _____	Answer __	☐ sure ☐ unsure ☐ guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

At a planning session ~~held in the school library~~, a teacher drafts questions for a data unit ~~that starts after winter break~~. Which is a statistical question?

Target word: identify a statistical question **Decides:** The answer must come from many values that vary.

A	Correct	Sleep varies from student to student; answering needs data from many.
B	Absolute language	Numeric answers can still be deterministic.
C	Adjacent category	One measurement with one answer; no variability is expected.
D	Confusion set	A single count, not a distribution.

2. Answer C Concept 2

A class ~~of 26~~ wonders whether students who play an instrument spend more time on homework. ~~The school has a new music wing.~~ What should the class do first?

Target word: first step of the cycle **Decides:** Every investigation starts by formulating the question.

A	Adjacent category	Analysis requires data that has not been collected yet.
B	Dropped element	Skips formulating; the class would not know exactly what to measure.
C	Correct	Formulating comes first and shapes what data to collect.
D	Inverted operation	Interpretation is the last step, not the first.

3. Answer D Concept 3

Heights of students measured ~~with a new stadiometer in the nurse's office~~ have a mean of 160 cm and a standard deviation of 8 cm. ~~The measurements took three days.~~ What is the z-score for a height of 176 cm?

Target word: z-score **Decides:** $(176 - 160) / 8$.

A	Wrong stem number	$176 \div 8$ skips subtracting the mean.
B	Dropped element	The raw difference, not divided by the SD.
C	Inverted operation	176 is above the mean, so z is positive.
D	Correct	$16 \div 8 = 2$ standard deviations above the mean.

4. Answer D Concept 3 · transfer

Scores on a test are normal with mean 500 and standard deviation 100. What score has $z = -1.5$?

Target word: solve for x **Decides:** $x = 500 + (-1.5)(100)$.

A	Confusion set	Adds z to the mean without multiplying by σ .
B	Inverted operation	Adds instead of subtracting for a negative z.
C	Dropped element	The distance from the mean, not the score.
D	Correct	$500 - 150 = 350$.

5. Answer C Concept 4

Two hospitals, ~~one of which has a new parking garage~~, record births: the larger has about 45 births a day and the smaller about 15. Over a year, which hospital has more days on which over 60% of babies born are boys?

Target word: sample size and variability **Decides:** Smaller daily samples vary more from 50%.

A	Absolute language	Daily proportions vary around 50%.
B	Confusion set	Equal probability does not mean equal day-to-day variability.
C	Correct	With 15 births a day, extreme proportions happen more often.
D	Inverted operation	Larger samples stay closer to the true proportion.

6. Answer A Concept 4 • transfer

A fair coin lands heads 7 times in its first 10 flips. As the total number of flips grows to 10,000, what is expected to happen?

Target word: long-run behavior **Decides:** Relative frequency approaches 0.5; past flips are not corrected.

A	Correct	The early surplus is swamped by thousands of new flips.
B	Dropped element	Ignores how the new flips dilute the first 10.
C	Adjacent category	The proportion converges; the raw difference can still grow.
D	Confusion set	The gambler's fallacy; the coin has no memory.

7. Answer D Concept 5

A principal wants a survey that represents grades 9 through 12, so she randomly selects 25 students from each grade. ~~The survey was emailed on a Monday.~~ What sampling method is this?

Target word: sampling method **Decides:** Random sampling happens within every group.

A	Camouflage	The delivery method does not change how students were selected.
B	Confusion set	Cluster sampling picks whole groups at random and surveys all members.
C	Adjacent category	An SRS draws from the whole school without guaranteeing each grade.
D	Correct	Each grade is a stratum, and students are randomly chosen within each one.

8. Answer B Concept 6

~~A student newspaper staff of 12~~ polls 400 randomly chosen students about a later start time. ~~The poll ran the week of the spring dance.~~ What is the approximate 95% margin of error?

Target word: margin of error from n **Decides:** $1/\sqrt{400} = 1/20$.

A	Dropped element	$1/400$ leaves out the square root.
B	Correct	$1/20 = 0.05$.
C	Confusion set	Halves the margin as if $\pm$ split it into two sides.
D	Inverted operation	Uses $\sqrt{400} = 20$ instead of $1/\sqrt{400}$.

9. Answer B Concept 6 • transfer

A pollster wants a 95% margin of error of about $\pm 2\%$ (an interval 4 points wide). About how many people should be sampled?

Target word: solve for n **Decides:** $1/\sqrt{n} = 0.02$, so $\sqrt{n} = 50$.

A	Wrong stem number	Uses the full 4-point width as the margin: $(1/0.04)^2$.
B	Correct	$\sqrt{n} = 50$, so $n = 2,500$.
C	Dropped element	That is $\sqrt{n}$; it still needs squaring.
D	Confusion set	Doubles 2,500 as if each side of $\pm$ needed its own sample.

10. Answer B Concept 7

At a school carnival that raised \$900, a game pays out if a player rolls at least one 6 in four rolls of a fair die. The booth was run by parent volunteers. What is the probability of winning?

Target word: P(at least one 6) **Decides:** $1 - (5/6)^4$.

A	Dropped element	$(5/6)^4$ is P(no 6); the $1 -$ step is missing.
B	Correct	$1 - 625/1296 \approx 0.518$.
C	Inverted operation	Adds $1/6$ four times instead of using the complement.
D	Confusion set	$(1/6)^4$ is the chance of four 6s.

11. Answer A Concept 7 • transfer

Three independent parts each have the same defect probability p . The chance that at least one is defective is 0.271. What is p ?

Target word: solve for p **Decides:** $(1 - p)^3 = 0.729$.

A	Correct	$\sqrt[3]{0.729} = 0.9$, so $p = 0.1$.
B	Adjacent category	$0.271 \div 3$ treats the events as simply additive.
C	Dropped element	Stops at P(no defects).
D	Inverted operation	That is $1 - p$, the chance a part is good.

12. Answer C Concept 8

During an after-lunch lesson, a student says HTHHTH is more likely than HHHHHH in six flips of a fair coin. The coins are plastic classroom counters. What is the correct response?

Target word: likelihood of specific sequences **Decides:** Each specific sequence has probability $(1/2)^6$.

A	Absolute language	It is unlikely but has the same $1/64$ chance.
B	Inverted operation	The hot-hand belief; flips are independent.
C	Correct	Each exact sequence of six fair flips has the same chance.
D	Confusion set	Mixes up “exactly three heads in any order” with one specific order.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	What makes a question statistical	A statistical question anticipates variability in the data and is answered by collecting and summarizing many values.	Thinking any question with a number in its answer is statistical.	“Which of these is a statistical question?”
2	The four-step investigation cycle	Formulate a question, collect data, analyze data, interpret results in context, in that order.	Starting with data or a graph before the question is set.	“What should the class do first?”
3	z-scores	$z = (x - \mu) / \sigma$ tells how many standard deviations a value is from the mean; $x = \mu + z\sigma$.	Dividing the raw value by σ , or forgetting the sign.	A mean, a standard deviation, and one value.
4	Law of large numbers	As the number of trials grows, the relative frequency settles near the true probability; small samples vary more.	Expecting outcomes to “balance out” in the short run, or thinking small and large samples are equally stable.	A comparison of small vs. large samples, or a streak followed by “what will happen?”
5	Stratified vs. cluster sampling	Stratified: divide into groups and randomly sample within every group; cluster: randomly pick whole groups and survey everyone in them.	Swapping the two because both use groups.	“From each grade” vs. “randomly chosen homerooms.”
6	Margin of error and sample size	For a 95% poll proportion, the margin of error is roughly $1/\sqrt{n}$, so cutting the margin in half needs four times the sample.	Using $1/n$, or thinking doubling n halves the margin.	A poll with n and “margin of error.”
7	“At least one” uses the complement	$P(\text{at least one}) = 1 - P(\text{none})$.	Adding probabilities across trials, or forgetting the $1 -$ step.	“At least one” across repeated independent trials.
8	The representativeness misconception	Every specific sequence of fair coin flips of the same length is equally likely, even if it does not “look random.”	Judging HTHHTH more likely than HHHHHH because it looks mixed.	Two specific sequences compared for likelihood.

Course scope (public description): Statistics for Secondary Mathematics Teaching provides a comprehensive exploration of advanced concepts and methodologies in teaching statistics and probability. The course delves into key topics such as summarizing and representing data, study design, sampling, testing claims, drawing conclusions, and probability. Emphasizing a deep understanding of conceptual foundations, including their historical development and perspectives from diverse cultures, the course addresses common misconceptions and students' ways of thinking. The appropriate use of instructional practices is also highlighted. There are no prerequisites for this course.